

# Aadi Desai

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## Profile

Software Engineer with experience building secure, performance-critical systems in Rust, and production-ready systems in Go and TypeScript. Record of driving development direction, stepping into technical leadership and mentoring engineers. Looking to develop infrastructure and systems-level design, and gain experience in full-stack, user-facing projects.

## Experience

### Software Engineer – Ultra Intelligence & Communications

Sep 2024 – Present

- Took over technical leadership of project direction following the departure of senior and principal engineers, including architecture decisions, milestone planning, and development prioritisation
- Mentored and supported engineers new to Rust, providing guidance on code quality, architecture design, and best practices for maintainable systems
- Implemented quantum-resistant cryptographic primitives with side-channel and memory safety assurances, with a focus on performance benchmarking and scaling on embedded hardware
- Built an async Rust watchdog service, including the diagnosis and resolution of cancel-safety issues around dropped futures as part of Axum's request handling
- Extended strategic proof-of-concept prototypes to stress-test and validate early-stage product opportunities, implementing hardware abstractions for new platforms
- Migrated service management from legacy init.d scripts to systemd service units as part of an in-house device orchestration framework, improving portability and operational reliability
- Built and maintained CI/CD pipelines for automated linting, testing, and release binary publishing
- Experimented with the ELK stack (Elasticsearch, Logstash, Kibana) for unified log visualisation
- Created a bus-based FPGA debug port, providing read/write access to user-defined registers within VHDL designs, for automated testing and programmatic access to signals in real-time

### Junior Embedded Software Engineer – Prolec Ltd

Apr 2024 – Aug 2024

- Worked in a close-knit team, engaging directly with customers and partner companies to gather requirements and incorporate feedback, with feature revisions built and delivered for same-day testing
- Optimised an embedded Windows application at the core of an automated safety system for construction and excavation equipment, including hazard geofencing and unsafe position limits
- Overhauled C++ codebase to reduce code size via conditional compilation and reduced class inheritance, enabling new features on space-constrained hardware
- Implemented the J1939 messaging protocol atop CAN bus, including safety-critical handshake confirmations for headless operation

### Digital Design Engineer – Nordic Semiconductor

Apr 2022 – Sep 2022

- Developed open-source editor tooling for SystemVerilog syntax highlighting and configurable linting, gathering feedback from colleagues to prioritise new linting rules and improvements
- Updated SystemVerilog simulation designs to use only synthesizable constructs, for compatibility with Verilator resulting in shorter functional-testing times and faster feedback loops

## Education

Imperial College London

*Oct 2019 – Jun 2023*

Master of Engineering in Electronics and Computer Engineering | First Class with Honours

## Skills

Languages: Rust, Go, SQL, TypeScript, Python, C++, SystemVerilog, VHDL, Solidity, F#

Tools & Infrastructure: Git, Linux, Docker, Cloudflare Workers, GitHub Actions

## Projects

**IRC-style Chat Server and Terminal Client** GitHub: [supleed2/go-chat](https://github.com/supleed2/go-chat)

*Dec 2023 – Jan 2024*

- Built a chat server and TUI client in Go following the Elm (MVU) architecture, communicating via a shared definition of a JSON WebSocket interface
- Server persists chat history across multiple rooms, using mutexes to prevent race conditions on shared mutable data, including client WebSockets and room state
- Containerised the application with Docker for easier setup and deployment as part of a wider multi-service Docker Compose home-lab environment
- Go learning project, including WebSockets, Goroutines and Channels for message passing

**Automated Verification Discord Bot** GitHub: [supleed2/nanobot](https://github.com/supleed2/nanobot)

*Jul 2023 – Aug 2023*

- Developed a Discord bot in Rust using the [Poise](#) framework for automated member onboarding, replacing manual verification that could be missed during busy periods
- Running since August 2023 with minimal downtime, with users providing feedback and feature requests
- Presents an Axum HTTP server for ingesting external validation of university records via Cloudflare Pages + Functions, avoiding storage of sensitive login information
- Migrated from the Shuttle platform with PostgreSQL to a generic container with SQLite and a reverse proxy, improving portability and eliminating platform dependency
- Open-source contributions: pull requests to upstream dependencies and tools used in this project, by forking and proposing changes

**Final Year Project: FPGA Accelerator for StackSynth** GitHub: [supleed2/EIE4-FYP](https://github.com/supleed2/EIE4-FYP)

*Oct 2022 – Jun 2023*

- [LiteX](#) + SystemVerilog project to generate audio, extending the Imperial 3<sup>rd</sup> Year Electrical Engineering Embedded Systems module with higher fidelity audio output
- LiteX wrapper module connects the custom designs and IP blocks, e.g. an integrated VexRiscV CPU
- Designed SystemVerilog modules which are instantiated in the LiteX wrapper: a CAN-bus receiver to accept messages from connected StackSynth boards, a 64-channel 48kHz configurable wave sample generator